

SOLIDWORKS API (.NET VBA C#) — Reference

(115/337): IFeatureManager (part 2 of 9)

SOLIDWORKS API Documentation (converted from CHM)

IFeatureManager (part 2 of 9) (Type)

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EditFreeze Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : EditFreeze Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::EditFreeze2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function EditFreeze( _  
    ByVal Location As System.Integer, _  
    ByVal FeatureName As System.String, _  
    ByVal UpdateAllConfigs As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Location As System.Integer  
Dim FeatureName As System.String  
Dim UpdateAllConfigs As System.Boolean  
Dim value As System.Integer  
  
value = instance.EditFreeze(Location, FeatureName, UpdateAllConfigs)
```

C#

```
System.int EditFreeze(  
    System.int Location,  
    System.string FeatureName,  
    System.bool UpdateAllConfigs  
)
```

C++/CLI

```
System.int EditFreeze(  
& System.int Location,  
& System.String^ FeatureName,  
& System.bool UpdateAllConfigs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Location

Location as defined in [swMoveFreezeBarTo_e](#)

FeatureName

Name of the feature

UpdateAllConfigs

True to update all configurations, false to not

Return Value

0 indicates that the freeze bar did not move to the specified location in the FeatureManager design tree; a non-0 value indicates that the freeze bar did move to the specified location in the FeatureManager design tree

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EditFreeze](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IFeatureManager::GetFreezeLocation Method](#)

[IModelDocExtension::NeedsRebuild2 Property](#)

[IModelDocExtension::UpdateFrozenFeatures Method](#)

[IFeature::HasFrozenUpdatePending Method](#)
[IFeature::IsFrozen Method](#)
[IFeature::MoveFreezeBarTo Method](#)
[IFeature::IsHiddenLock Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

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EditFreeze2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager interface](#) : EditFreeze2 Method (IFeatureManager)

Moves the freeze bar to the specified location in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Function EditFreeze2( _  
    ByVal Location As System.Integer, _  
    ByVal FeatureName As System.String, _  
    ByVal UpdateAllConfigs As System.Boolean, _  
    ByVal UnlockConfigs As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Location As System.Integer  
Dim FeatureName As System.String  
Dim UpdateAllConfigs As System.Boolean  
Dim UnlockConfigs As System.Boolean  
Dim value As System.Integer  
  
value = instance.EditFreeze2(Location, FeatureName, UpdateAllConfigs, UnlockConfigs)
```

C#

```
System.int EditFreeze2(  
    System.int Location,  
    System.string FeatureName,  
    System.bool UpdateAllConfigs,  
    System.bool UnlockConfigs  
)
```

C++/CLI

```
System.int EditFreeze2(  
& System.int Location,  
& System.String^ FeatureName,  
& System.bool UpdateAllConfigs,  
& System.bool UnlockConfigs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Location

Location as defined in [swMoveFreezeBarTo e](#)

FeatureName

Name of the feature

UpdateAllConfigs

True to update all configurations, false to not

UnlockConfigs

True to unlock configurations, false to not

Return Value

1 indicates that the freeze bar did not move to the specified location in the FeatureManager design tree; 0 indicates that the freeze bar did move to the specified location in the FeatureManager design tree

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EditFreeze2](#).

Example

[Move Freeze Bar \(VBA\)](#)
[Move Freeze Bar \(VB.NET\)](#)
[Move Freeze Bar \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::GetFreezeLocation Method \(\)](#)
[IFeature::HasFrozenUpdatePending Method \(\)](#)
[IFeature::IsFrozen Method \(\)](#)
[IFeature::MoveFreezeBarTo2 Method \(\)](#)
[IFeature::IsHiddenLock Method \(\)](#)
[IModelDocExtension::NeedsRebuild2 Property \(\)](#)
[IModelDocExtension::UpdateFrozenFeatures Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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EditReferencePoint Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : EditReferencePoint Method (IFeatureManager)

Edits the selected reference points.

.NET Syntax

Visual Basic (Declaration)

```
Function EditReferencePoint( _  
    ByVal NRefPointType As System.Integer, _  
    ByVal NRefPointAlongCurveType As System.Integer, _  
    ByVal DDistance\_or\_Percent As System.Double, _  
    ByVal NumberOfRefPoints As System.Integer _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim NRefPointType As System.Integer  
Dim NRefPointAlongCurveType As System.Integer  
Dim DDistance_or_Percent As System.Double  
Dim NumberOfRefPoints As System.Integer  
Dim value As System.Boolean
```

```
value = instance.EditReferencePoint(NRefPointType, NRefPointAlongCurveType, DDistance_or_Percent,  
NumberOfRefPoints)
```

C#

```
System.bool EditReferencePoint(  
    System.int NRefPointType,  
    System.int NRefPointAlongCurveType,  
    System.double DDistance\_or\_Percent,  
    System.int NumberOfRefPoints  
)
```

C++/CLI

```
System.bool EditReferencePoint(  
& System.int NRefPointType,  
& System.int NRefPointAlongCurveType,
```

```
& System.double DDistance or Percent,
& System.int NumberOfRefPoints
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NRefPointType

Type of reference point as defined by [swRefPointType_e](#)

NRefPointAlongCurveType

Distance, percentage, or evenly distributed as defined by [swRefPointAlongCurveType_e](#)

DDistance_or_Percent

Distance at which to create the reference point on the selected entities or percentage of the length of the selected entities at which to create the reference point if *NRefPointAlongCurveType* is *swRefPointAlongCurveDistance* or *swRefPointAlongCurvePercentage*, respectively

NumberOfRefPoints

Number of reference points to create and evenly distribute on the selected entities if *swRefPointAlongCurveType* is *swRefPointAlongCurveEvenlyDistributed*

Return Value

True if the operation succeeds, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EditReferencePoint](#).

Remarks

A reference point is a feature. To programatically create a reference point feature, you can use [IFeatureManager::InsertReferencePoint](#) or [IFeatureManager::InsertReferencePoint](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IRefPointFeatureData Interface](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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EditRollback Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : EditRollback Method (IFeatureManager)

Rolls back or forward the rollback bar to a specific location in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Function EditRollback( _
    ByVal Location As System.Integer, _
    ByVal Feature As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Location As System.Integer
Dim Feature As System.String
Dim value As System.Boolean
```

```
value = instance.EditRollback(Location, Feature)
```

C#

```
System.bool EditRollback(
    System.int Location,
    System.string Feature
)
```

C++/CLI

```
System.bool EditRollback(  
& System.int Location,  
& System.String^ Feature  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Location

Location to which to move the rollback bar as defined in [swMoveRollbackBarTo_e](#)

Feature

Name of the feature

Return Value

True if moving the rollback bar back or forward was successful, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EditRollback](#).

Example

[Roll Back Model \(C#\)](#)
[Roll Back Model \(VB.NET\)](#)
[Roll Back Model \(VBA\)](#)

Remarks

When the model is in a rollback state, it reverts to an earlier state, suppressing recently added features. You can add new features or edit existing features while the model is in the rolled-back state.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeature::IsRolledBack Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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EnableFeatureTree Property (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : EnableFeatureTree Property (IFeatureManager)

Gets or sets whether or not to update the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

Property EnableFeatureTree As System.Boolean

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Boolean
```

```
instance.EnableFeatureTree = value
```

```
value = instance.EnableFeatureTree
```

C#

```
System.bool EnableFeatureTree {get; set;}
```

C++/CLI

```
property System.bool EnableFeatureTree {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to update FeatureManager design tree, false to not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EnableFeatureTree](#).

Example

[Only Show Selected Components \(VBA\)](#)

Remarks

Use this property to temporarily enable or disable the FeatureManager design tree while an add-in is running a series of operations. Temporarily disabling the FeatureManager design tree should increase performance. However, while the FeatureManager design tree is disabled and not being updated, it may become out of date with what is actually in the model.

If an interactive user attempts to interact with the FeatureManager design tree while it is in this out-of-date state, problems could occur. Normally this should not happen because the add-in should only be disabling the FeatureManager design tree update temporarily and then enabling it immediately after the series of operations completes. However, if there is any possibility that the interactive user could interact with the FeatureManager design tree while it is an out-of-date state, then also use [IFeatureManager::EnableFeatureTreeWindow](#), which prohibits any user interaction with the FeatureManager design tree because the control itself is disabled.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::UpdateFeatureTree Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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EnableFeatureTreeWindow Property (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : EnableFeatureTreeWindow Property (IFeatureManager)

Gets or sets whether the FeatureManager design tree is enabled or not.

.NET Syntax

Visual Basic (Declaration)

```
Property EnableFeatureTreeWindow As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Boolean
```

```
instance.EnableFeatureTreeWindow = value
```

```
value = instance.EnableFeatureTreeWindow
```

C#

```
System.bool EnableFeatureTreeWindow {get; set;}
```

C++/CLI

```
property System.bool EnableFeatureTreeWindow {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EnableFeatureTreeWindow](#).

Remarks

Use this property with [IFeatureManager::EnableFeatureTree](#), which temporarily enables or disables the FeatureManager design tree update when an add-in is running a series of operations.

While the FeatureManager design tree is disabled and not being updated, the FeatureManager tree may become out of date with what is actual in the model. If an interactive user attempts to interact with the FeatureManager design tree while it is in this out-of-date state, problems could occur. Normally, this should not happen because the add-in should only be disabling the FeatureManager design tree update temporarily and then enabling it immediately after the series of operations complete. However, if there is any possibility that an interactive user could interact with the FeatureManager design tree, then use [FeatureManager::EnableFeatureTreeWindow](#), which prohibits any user interaction with the FeatureManager design tree because the control is disabled.

Just as with the [IFeatureManager::EnableFeatureTree](#) property, this property is meant to be used for temporary disabling. If it is necessary to disable the FeatureManager design tree control, then it should only be disabled while the add-in is performing some series of operations and enabled immediately after these operations complete. Do not to leave the FeatureManager design tree window disabled when control is returned to SOLIDWORKS, i.e., the interactive user.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::UpdateFeatureTree Method](#)

Availability

SOLIDWORKS 2005 SP2, Revision Number 13.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

EndVariablePitchHelix Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : EndVariablePitchHelix Method (IFeatureManager)

Ends and inserts a variable-pitch helix.

.NET Syntax

Visual Basic (Declaration)

```
Function EndVariablePitchHelix() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As Feature
```

```
value = instance.EndVariablePitchHelix()
```

C#

```
Feature EndVariablePitchHelix()
```

C++/CLI

```
Feature^ EndVariablePitchHelix();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EndVariablePitchHelix](#).

Example

[Create Variable-pitch Helix \(C#\)](#)
[Create Variable-pitch Helix \(VB.NET\)](#)
[Create Variable-pitch Helix \(VBA\)](#)
[Create and Modify Variable-pitch Helix \(C#\)](#)
[Create and Modify Variable-pitch Helix \(VB.NET\)](#)
[Create and Modify Variable-pitch Helix \(VBA\)](#)

Remarks

To create and insert a variable-pitch helix, call:

1. [IFeatureManager::InsertVariablePitchHelix](#) to create it.
2. [IFeatureManager::AddVariablePitchHelixFirstPitchAndDiameter](#) to add the first segment.
3. [IFeatureManager::AddVariablePitchHelixSegment](#) one or more times to add the remaining segments.
4. [IFeatureManager::EndVariablePitchHelix](#) to insert it.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IHelixFeatureData Interface](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ExpandFeature Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : ExpandFeature Method (IFeatureManager)

Expands the specified component in the specified FeatureManager design tree pane.

.NET Syntax

Visual Basic (Declaration)

```
Function ExpandFeature( _  
    ByVal Component As System.Object, _  
    ByVal FeatureName As System.String, _  
    ByVal WhichPane As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Component As System.Object  
Dim FeatureName As System.String  
Dim WhichPane As System.Integer  
Dim value As System.Boolean
```

```
value = instance.ExpandFeature(Component, FeatureName, WhichPane)
```

C#

```
System.bool ExpandFeature(  
    System.object Component,  
    System.string FeatureName,  
    System.int WhichPane  
)
```

C++/CLI

```
System.bool ExpandFeature(
& System.Object^ Component,
& System.String^ FeatureName,
& System.int WhichPane
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Component
[Component](#)
FeatureName
Name of a feature in Component
WhichPane
FeatureManager design tree pane in which to expand Component as defined in [swFeatMgrPane_e](#)

Return Value

True if the specified component is expanded in the specified FeatureManager design tree pane, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::ExpandFeature](#).

Example

[Expand Component in Specified FeatureManager Design Tree Pane \(C#\)](#)
[Expand Component in Specified FeatureManager Design Tree Pane \(VB.NET\)](#)
[Expand Component in Specified FeatureManager Design Tree Pane \(VBA\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ITreeControlItem::Expanded Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureAdvancedTableDrivenPattern Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureAdvancedTableDrivenPattern Method (IFeatureManager)

Obsolete. See [IFeatureManager::CreateFeature](#) and the Remarks in [IDimPatternFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureAdvancedTableDrivenPattern() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As Feature
```

```
value = instance.FeatureAdvancedTableDrivenPattern()
```

C#

```
Feature FeatureAdvancedTableDrivenPattern()
```

C++/CLI

```
Feature^ FeatureAdvancedTableDrivenPattern();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureAdvancedTableDrivenPattern](#).

Remarks

This method inserts a variable pattern feature, which uses a table to store the dimensions and values of the pattern instances.
Call this method after calling [IFeatureManager::InsertVaryInstanceOverride](#) to create an advanced variable pattern feature.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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FeatureBossThicken Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureBossThicken Method (IFeatureManager)

Thickens the selected reference surface, and then generates a boss.

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureBossThicken( _  
    ByVal Thickness As System.Double, _  
    ByVal Direction As System.Integer, _  
    ByVal FaceIndex As System.Integer, _  
    ByVal FillVolume As System.Boolean, _  
    ByVal Merge As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Thickness As System.Double  
Dim Direction As System.Integer  
Dim FaceIndex As System.Integer  
Dim FillVolume As System.Boolean  
Dim Merge As System.Boolean  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean  
Dim value As Feature
```

```
value = instance.FeatureBossThicken(Thickness, Direction, FaceIndex, FillVolume, Merge,  
UseFeatScope, UseAutoSelect)
```

C#

```
Feature FeatureBossThicken(  
    System.double Thickness,  
    System.int Direction,  
    System.int FaceIndex,  
    System.bool FillVolume,  
    System.bool Merge,  
    System.bool UseFeatScope,
```

```
        System.bool UseAutoSelect
    )
```

C++/CLI

```
Feature^ FeatureBossThicken(
&    System.double Thickness,
&    System.int Direction,
&    System.int FaceIndex,
&    System.bool FillVolume,
&    System.bool Merge,
&    System.bool UseFeatScope,
&    System.bool UseAutoSelect
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness

Wall thickness

Direction

Direction as defined in [swThickenThicknessType_e](#)

FaceIndex

Not used

FillVolume

True if you want to make a solid from the knitted surface, false if not (see **Remarks**)

Merge

True if you want to merge the results in a multibody part, false if not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureBossThicken](#).

Example

[Create Thicken Feature \(VBA\)](#)

[Fill Holes in Part \(C#\)](#)

[Fill Holes in Part \(VB.NET\)](#)

[Fill Holes in Part \(VBA\)](#)

[Thicken Surface and Generate Boss \(VBA\)](#)

[Thicken Surface and Generate Boss \(VB.NET\)](#)

[Thicken Surface and Generate Boss \(C#\)](#)

Remarks

This method generates a boss feature by thickening a selected reference surface. The surface must be selected with a mark of 1. See [IModelDocExtension::SelectById2](#) for details.

If FillVolume is true, other arguments are ignored. A closed surface is required when FillVolume is true.

If UseAutoSelect is false, the user must select the bodies that the feature will affect.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IThickenFeatureData Interface](#)

[IFeatureManager::FeatureCutThicken Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0



Obsolete. See [IFeatureManager::CreateFeature](#) and the Remarks in [IChainPatternFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureChainPattern( _  
    ByVal PitchMethod As System.Integer, _  
    ByVal FlipDirection As System.Boolean, _  
    ByVal FillPath As System.Boolean, _  
    ByVal Number As System.Integer, _  
    ByVal Spacing As System.Double, _  
    ByVal GroupOneFlipPlane As System.Boolean, _  
    ByVal GroupTwoChain As System.Boolean, _  
    ByVal GroupTwoFlipPlane As System.Boolean, _  
    ByVal AlignMethod As System.Integer, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim PitchMethod As System.Integer  
Dim FlipDirection As System.Boolean  
Dim FillPath As System.Boolean  
Dim Number As System.Integer  
Dim Spacing As System.Double  
Dim GroupOneFlipPlane As System.Boolean  
Dim GroupTwoChain As System.Boolean  
Dim GroupTwoFlipPlane As System.Boolean  
Dim AlignMethod As System.Integer  
Dim Options As System.Integer  
Dim value As Feature
```

```
value = instance.FeatureChainPattern(PitchMethod, FlipDirection, FillPath, Number, Spacing,  
GroupOneFlipPlane, GroupTwoChain, GroupTwoFlipPlane, AlignMethod, Options)
```

C#

```
Feature FeatureChainPattern(  
    System.int PitchMethod,  
    System.bool FlipDirection,  
    System.bool FillPath,  
    System.int Number,  
    System.double Spacing,  
    System.bool GroupOneFlipPlane,  
    System.bool GroupTwoChain,  
    System.bool GroupTwoFlipPlane,  
    System.int AlignMethod,  
    System.int Options  
)
```

C++/CLI

```
Feature^ FeatureChainPattern(  
& System.int PitchMethod,  
& System.bool FlipDirection,  
& System.bool FillPath,  
& System.int Number,  
& System.double Spacing,  
& System.bool GroupOneFlipPlane,  
& System.bool GroupTwoChain,  
& System.bool GroupTwoFlipPlane,  
& System.int AlignMethod,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PitchMethod
Pitch method as defined in [swChainPatternPitchMethod_e](#); available for all types of chain patterns

FlipDirection
True to reverse the direction of the chain pattern, false to not; available for all types of chain patterns

FillPath
True to automatically set the number of pattern instances to fill the path, false to not; available for all types of chain patterns and when EqualSpacing is false for distance and linkage distance chain patterns

Number
Number of pattern instances; available for all types of chain patterns when FillPath is false

Spacing
Distance between the pattern instances when EqualSpacing is false; only available for distance and linkage distance chain patterns when EqualSpacing is false

GroupOneFlipPlane
True to flip the path alignment plane for **Chain Group 1**, false to not; available for all types of chain patterns

GroupTwoChain
True to set the component to pattern for **Chain Group 2**, false to not; only available for connected linkage chain patterns

GroupTwoFlipPlane
True to flip the path alignment plane for **Chain Group 2**, false to not; only available for connected linkage chain patterns

AlignMethod
Alignment method as defined in [swChainPatternAlignment_e](#); only available for distance and linkage distance chain patterns

Options
Option as defined in [swChainPatternOptions_e](#); available for all types of chain patterns

Return Value

Chain pattern [feature](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureChainPattern](#).

 Remarks

You must pre-select the entities for the chain pattern feature. Using the selected entities, this method inserts one of these chain pattern features:

- distance
- distance linkage
- connected linkage

Entity to select	Corresponding Chain Pattern PropertyManager control	Selection mark	Number of selections
• 2D or 3D sketch containing an open or closed loop, • Sketch line, or • Model edge	Path	2	1 for all types of chain patterns
Assembly component	Chain Group 1 Component to Pattern	1	1 for all types of chain patterns
• Cylindrical face, • Circular or linear edge, • Sketch point, • Vertex, or • Reference axis	Chain Group 1 Path Link 1	256	1 for all types of chain patterns
• Cylindrical face, • Circular or linear edge, • Sketch point, • Vertex, or • Reference axis	Chain Group 1 Path Link 2	512	• 1 for distance linkage or connected linkage • None for distance
Component plane or planar face	Chain Group 1 Path Alignment Plane	16384	1 for all types of chain patterns
Assembly component	Chain Group 2 Component to Pattern	2048	• 1 for connected linkage • None for distance or linkage distance
• Cylindrical face, • Circular or linear edge, • Sketch point, • Vertex, or • Reference axis	Chain Group 2 Path Link 1	4096	• 1 for connected linkage • None for distance or linkage distance
• Cylindrical face, • Circular or linear edge, • Sketch point, • Vertex, or • Reference axis	Chain Group 2 Path Link 2	8192	• 1 for connected linkage • None for distance or linkage distance
Component plane or planar face	Group 2 Path Alignment Plane	32768	• 1 for connected linkage • None for distance or linkage distance
Assembly plane	Face normal alignment	1024	• 1 if the chain path is a sketch line • None for all other types of paths

To [set equal spacing](#) between chain pattern instances, edit the chain pattern feature after creating it.

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

 Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCircularPattern Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureCircularPattern2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCircularPattern( _  
    ByVal Num As System.Integer, _  
    ByVal Spacing As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal DName As System.String _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Num As System.Integer  
Dim Spacing As System.Double  
Dim FlipDir As System.Boolean  
Dim DName As System.String  
Dim value As Feature  
  
value = instance.FeatureCircularPattern(Num, Spacing, FlipDir, DName)
```

C#

```
Feature FeatureCircularPattern(  
    System.int Num,  
    System.double Spacing,  
    System.bool FlipDir,  
    System.string DName  
)
```

C++/CLI

```
Feature^ FeatureCircularPattern(  
& System.int Num,  
& System.double Spacing,  
& System.bool FlipDir,  
& System.String^ DName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Num

Spacing

FlipDir

DName

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCircularPattern](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)



Obsolete. Superseded by [IFeatureManager::FeatureCircularPattern3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCircularPattern2( _
    ByVal Num As System.Integer, _
    ByVal Spacing As System.Double, _
    ByVal FlipDir As System.Boolean, _
    ByVal DName As System.String, _
    ByVal GeometryPattern As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Num As System.Integer
Dim Spacing As System.Double
Dim FlipDir As System.Boolean
Dim DName As System.String
Dim GeometryPattern As System.Boolean
Dim value As Feature

value = instance.FeatureCircularPattern2(Num, Spacing, FlipDir, DName, GeometryPattern)
```

C#

```
Feature FeatureCircularPattern2(
    System.int Num,
    System.double Spacing,
    System.bool FlipDir,
    System.string DName,
    System.bool GeometryPattern
)
```

C++/CLI

```
Feature^ FeatureCircularPattern2(
    & System.int Num,
    & System.double Spacing,
    & System.bool FlipDir,
    & System.String^ DName,
    & System.bool GeometryPattern
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Num
Number of instances of the circular pattern to insert, including the original

Spacing
Spacing between each instance of the circular pattern in radians

FlipDir
True to flip the direction of the circular pattern, false to not

DName
Name of the angular dimension defining the direction of the pattern

GeometryPattern
True to use geometry pattern, false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCircularPattern2](#).

Remarks

This method requires ordered selection of the features and components.

- Features and axis: Use [IModelDocExtension::SelectByID2](#) with a Mark of 4 for the features to pattern and a Mark of 1 for the axis. You can also use [IFeatureManager::Select2](#) to select the features and the axis.
- Components and axis: Use [IModelDocExtension::SelectByID2](#) with a Mark of 1 for the components to pattern and a Mark of 2 for the axis.

You must select the features or components before selecting the axis.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ICircularPatternFeatureData Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCircularPattern3 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCircularPattern3 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureCircularPattern4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCircularPattern3( _
    ByVal Number As System.Integer, _
    ByVal Spacing As System.Double, _
    ByVal FlipDirection As System.Boolean, _
    ByVal DName As System.String, _
    ByVal GeometryPattern As System.Boolean, _
    ByVal EqualSpacing As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Number As System.Integer
Dim Spacing As System.Double
Dim FlipDirection As System.Boolean
Dim DName As System.String
Dim GeometryPattern As System.Boolean
Dim EqualSpacing As System.Boolean
Dim value As Feature
```

```
value = instance.FeatureCircularPattern3(Number, Spacing, FlipDirection, DName, GeometryPattern, EqualSpacing)
```

C#

```
Feature FeatureCircularPattern3(
    System.int Number,
    System.double Spacing,
    System.bool FlipDirection,
    System.string DName,
    System.bool GeometryPattern,
    System.bool EqualSpacing
)
```

C++/CLI

```
Feature^ FeatureCircularPattern3(
    & System.int Number,
    & System.double Spacing,
    & System.bool FlipDirection,
    & System.String^ DName,
    & System.bool GeometryPattern,
```

```
& System.bool EqualSpacing  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Number
Number of instances of the circular pattern to insert, including the original instance
Spacing
Spacing between each instance of the circular pattern or total angle if EqualSpacing is true (in radians)
FlipDirection
True to flip the direction of the circular pattern, false to not
DName
Name of the angular dimension defining the direction of the pattern
GeometryPattern
True to use geometry pattern, false to not
EqualSpacing
True to use equal spacing, false to not

Return Value

Circular pattern [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCircularPattern3](#).

Remarks

This method requires ordered selection of the features and components.

- Features and axis: Use [IModelDocExtension::SelectByID2](#) with a Mark of 4 for the features to pattern and a Mark of 1 for the axis. You can also use [IFeature::Select2](#) to select the features and the axis.
- Components and axis: Use IModelDocExtension::SelectByID2 with a Mark of 1 for the components to pattern and a Mark of 2 for the axis.

You must select the features or components before selecting the axis.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ICircularPatternFeatureData Interface](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCircularPattern4 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCircularPattern4 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureCircularPattern5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCircularPattern4( _  
    ByVal Number As System.Integer, _  
    ByVal Spacing As System.Double, _  
    ByVal FlipDirection As System.Boolean, _  
    ByVal DName As System.String, _  
    ByVal GeometryPattern As System.Boolean, _  
    ByVal EqualSpacing As System.Boolean, _  
    ByVal VaryInstance As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Number As System.Integer  
Dim Spacing As System.Double
```



```
Dim FlipDirection As System.Boolean
Dim DName As System.String
Dim GeometryPattern As System.Boolean
Dim EqualSpacing As System.Boolean
Dim VaryInstance As System.Boolean
Dim value As Feature
```

```
value = instance.FeatureCircularPattern4(Number, Spacing, FlipDirection, DName, GeometryPattern,
EqualSpacing, VaryInstance)
```

C#

```
Feature FeatureCircularPattern4(
    System.int Number,
    System.double Spacing,
    System.bool FlipDirection,
    System.string DName,
    System.bool GeometryPattern,
    System.bool EqualSpacing,
    System.bool VaryInstance
)
```

C++/CLI

```
Feature^ FeatureCircularPattern4(
    & System.int Number,
    & System.double Spacing,
    & System.bool FlipDirection,
    & System.String^ DName,
    & System.bool GeometryPattern,
    & System.bool EqualSpacing,
    & System.bool VaryInstance
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Number

Number of instances of the circular pattern to insert, including the original instance

Spacing

Spacing between each instance of the circular pattern; total angle in radians if EqualSpacing is true

FlipDirection

True to flip the direction of the circular pattern, false to not

DName

Name of the angular dimension defining the direction of the pattern

GeometryPattern

True to use geometry pattern, false to not

EqualSpacing

True to use equal spacing, false to not

VaryInstance

True to vary the dimensions or spacing of individual pattern instances, false to not; valid only if GeometryPattern = false (see **Remarks**)

Return Value

Circular pattern [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.FeatureCircularPattern4](#).

Example

[Create Circular Pattern \(VBA\)](#)

[Create Circular Pattern \(VB.NET\)](#)

[Create Circular Pattern \(C#\)](#)

Remarks

If...	To select a feature, use...	To select a component, use...
Using IModelDocExtension::SelectByID2 to select features and components, ordered selection of the features and components is required	1 to mark the direction axis 4 to mark the features to pattern	1 to mark the components to pattern 2 to mark the direction axis

Directly selecting a feature or axis using
[IFeatureManager::Select2](#)

- 1 to mark the direction axis
- 4 to mark the features to pattern

[ISelectData::Mark](#) and
[Component2::Select3](#)

- 1 to mark the components to pattern
- 2 to mark the direction axis

If VaryInstance = true, then to indicate how to vary the individual pattern instances, decide on the type of pattern and call its corresponding method before calling this method:

Type...	Method...
Increment	IFeatureManager::InsertVaryInstanceIncrement
Override	IFeatureManager::InsertVaryInstanceOverride

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ICircularPatternFeatureData Interface](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCircularPattern5 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCircularPattern5 Method (IFeatureManager)

Obsolete. See [IFeatureManager::CreateFeature](#) and the Remarks in [ICircularPatternFeatureData](#) and [ILocalCircularPatternFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCircularPattern5( _  
    ByVal Number As System.Integer, _  
    ByVal Spacing As System.Double, _  
    ByVal FlipDirection As System.Boolean, _  
    ByVal DName As System.String, _  
    ByVal GeometryPattern As System.Boolean, _  
    ByVal EqualSpacing As System.Boolean, _  
    ByVal VaryInstance As System.Boolean, _  
    ByVal SyncSubAssemblies As System.Boolean, _  
    ByVal BDir2 As System.Boolean, _  
    ByVal BSymmetric As System.Boolean, _  
    ByVal Number2 As System.Integer, _  
    ByVal Spacing2 As System.Double, _  
    ByVal DName2 As System.String, _  
    ByVal EqualSpacing2 As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Number As System.Integer  
Dim Spacing As System.Double  
Dim FlipDirection As System.Boolean  
Dim DName As System.String  
Dim GeometryPattern As System.Boolean  
Dim EqualSpacing As System.Boolean  
Dim VaryInstance As System.Boolean  
Dim SyncSubAssemblies As System.Boolean  
Dim BDir2 As System.Boolean  
Dim BSymmetric As System.Boolean  
Dim Number2 As System.Integer  
Dim Spacing2 As System.Double  
Dim DName2 As System.String
```



```
Dim EqualSpacing2 As System.Boolean
Dim value As Feature
```

```
value = instance.FeatureCircularPattern5(Number, Spacing, FlipDirection, DName, GeometryPattern,
EqualSpacing, VaryInstance, SyncSubAssemblies, BDir2, BSymmetric, Number2, Spacing2, DName2,
EqualSpacing2)
```

C#

```
Feature FeatureCircularPattern5(
    System.int Number,
    System.double Spacing,
    System.bool FlipDirection,
    System.string DName,
    System.bool GeometryPattern,
    System.bool EqualSpacing,
    System.bool VaryInstance,
    System.bool SyncSubAssemblies,
    System.bool BDir2,
    System.bool BSymmetric,
    System.int Number2,
    System.double Spacing2,
    System.string DName2,
    System.bool EqualSpacing2
)
```

C++/CLI

```
Feature^ FeatureCircularPattern5(
    & System.int Number,
    & System.double Spacing,
    & System.bool FlipDirection,
    & System.String^ DName,
    & System.bool GeometryPattern,
    & System.bool EqualSpacing,
    & System.bool VaryInstance,
    & System.bool SyncSubAssemblies,
    & System.bool BDir2,
    & System.bool BSymmetric,
    & System.int Number2,
    & System.double Spacing2,
    & System.String^ DName2,
    & System.bool EqualSpacing2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Number

Number of instances of the circular pattern to insert in Direction 1, including the original instance

Spacing

Spacing between each instance in Direction 1 of the circular pattern or, if EqualSpacing is true, then the total angle in radians

FlipDirection

True to flip the direction of the circular pattern in Direction 1, false to not

DName

Name of the angular dimension defining Direction 1 of the pattern

GeometryPattern

True to use geometry pattern, false to not

EqualSpacing

True to use equal spacing in Direction 1, false to not

VaryInstance

True to vary the dimensions or spacing of individual pattern instances, false to not; valid only if GeometryPattern = false (see **Remarks**)

SyncSubAssemblies

True to move components in the patterned instances when components are moved in the seed flexible subassembly, false to not

BDir2

True to create a bidirectional circular pattern feature, false to not

BSymmetric

True to create a symmetric circular pattern feature in Direction 2, false to create an asymmetrical circular pattern feature in Direction 2; valid only if BDir2 is true

Number2

Number of instances to insert in Direction 2; valid only if BDir2 is true

Spacing2

Distance between pattern instances in Direction 2; valid only if BDir2 is true

DName2

Name of the angular dimension defining Direction 2 of the pattern; valid only if BDir2 is true

EqualSpacing2

True to use equal spacing in Direction 2, false to not; valid only if BDir2 is true and BSymmetric is false

Return Value

Circular pattern [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCircularPattern5](#).

Remarks

If...	To select a feature, use...	To select a component, use...
Using IModelDocExtension::SelectByID2 to select features and components, ordered selection of the features and components is required	• 1 to mark the direction axis • 4 to mark the features to pattern	• 1 to mark the components to pattern • 2 to mark the direction axis
Directly selecting a feature or axis using IFeature::Select2	• 1 to mark the direction axis • 4 to mark the features to pattern	ISelectData::Mark and Component2::Select3 • 1 to mark the components to pattern • 2 to mark the direction axis

If VaryInstance = true, then to indicate how to vary the individual pattern instances, decide on the type of pattern and call its corresponding method before calling this method:

Type	Method
Increment	IFeatureManager::InsertVaryInstanceIncrement
Override	IFeatureManager::InsertVaryInstanceOverride

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCut Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCut Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureCut2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCut( _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean, _  
    ByVal TranslateSurface1 As System.Boolean, _  
    ByVal TranslateSurface2 As System.Boolean, _
```

```

        ByVal NormalCut As System.Boolean, _
        ByVal UseFeatScope As System.Boolean, _
        ByVal UseAutoSelect As System.Boolean _
    ) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim NormalCut As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim value As Feature

```

```

value = instance.FeatureCut(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, NormalCut,
UseFeatScope, UseAutoSelect)

```

C#

```

Feature FeatureCut(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool NormalCut,
    System.bool UseFeatScope,
    System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ FeatureCut(
    & System.bool Sd,
    & System.bool Flip,
    & System.bool Dir,
    & System.int T1,
    & System.int T2,
    & System.double D1,
    & System.double D2,
    & System.bool Dchk1,

```

```

& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool NormalCut,
& System.bool UseFeatScope,
& System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single-ended, false for doubled-ended

Flip

Flip to side cut; true if you want to remove material outside of the profile, false if not

Dir

Reverse direction; true if you want Direction 1 to be opposite of the default direction (see **Remarks**)

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

True for first draft angle to be inward, false to be outward

Ddir2

True for second draft angle to be inward, false be outward

Dang1

Draft angle for first end

Dang2

Draft angle for second end

OffsetReverse1

If you chose to offset the first end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

OffsetReverse2

If you chose to offset the second end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

TranslateSurface1

When you choose `swEndcondOffsetFromSurface` as the termination type for the first end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

TranslateSurface2

When you choose `swEndcondOffsetFromSurface` as the termination type for the second end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

NormalCut

True ensures that the cut is created normal to the sheet metal thickness, false does not (this argument applies only to sheet metal parts, use false for non-sheet metal parts)

UseFeatScope

True if the feature only affects selected bodies or components, false if the feature affects all bodies or components

UseAutoSelect

True to automatically select all bodies or components and have the feature affect those bodies or components, false to select the bodies or components the feature affects (see **Remarks**)

Return Value

Pointer to the [IFeature](#) object

Remarks

The default direction for cut operations are opposite the sketch normal. The default direction for boss operations is along the sketch normal. Setting the direction arguments to true reverse the default direction. For double-ended extrusions, Direction 2 is always be opposite to Direction 1.

The default sketch normal is the same as the face or plane normal where the sketch was placed. To determine this normal vector, see [IFace2::Normal](#) and [IRefPlane::Transform](#), respectively.

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::FeatureCutThicken Method](#)
[IFeatureManager::FeatureCutThin Method](#)
[IExtrudeFeatureData2 Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCut2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCut2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureCut3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCut2( _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean, _  
    ByVal TranslateSurface1 As System.Boolean, _  
    ByVal TranslateSurface2 As System.Boolean, _  
    ByVal NormalCut As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean, _  
    ByVal AssemblyFeatureScope As System.Boolean, _  
    ByVal AutoSelectComponents As System.Boolean, _  
    ByVal PropagateFeatureToParts As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Sd As System.Boolean  
Dim Flip As System.Boolean  
Dim Dir As System.Boolean
```



```

Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim NormalCut As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim PropagateFeatureToParts As System.Boolean
Dim value As Feature

```

```

value = instance.FeatureCut2(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, NormalCut,
UseFeatScope, UseAutoSelect, AssemblyFeatureScope, AutoSelectComponents, PropagateFeatureToParts)

```

C#

```

Feature FeatureCut2(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool NormalCut,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.bool AssemblyFeatureScope,
    System.bool AutoSelectComponents,
    System.bool PropagateFeatureToParts
)

```

C++/CLI

```

Feature^ FeatureCut2(
    & System.bool Sd,
    & System.bool Flip,
    & System.bool Dir,
    & System.int T1,
    & System.int T2,
    & System.double D1,
    & System.double D2,
    & System.bool Dchk1,
    & System.bool Dchk2,
    & System.bool Ddir1,
    & System.bool Ddir2,
    & System.double Dang1,
    & System.double Dang2,

```

```

& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool NormalCut,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents,
& System.bool PropagateFeatureToParts
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single-ended, false for doubled-ended

Flip

Flip to side cut; true if you want to remove material outside of the profile, false if not

Dir

Reverse direction; true if you want Direction 1 to be opposite of the default direction (see **Remarks**)

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

True for first draft angle to be inward, false to be outward

Ddir2

True for second draft angle to be inward, false be outward

Dang1

Draft angle for first end

Dang2

Draft angle for second end

OffsetReverse1

If you chose to offset the first end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

OffsetReverse2

If you chose to offset the second end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

TranslateSurface1

When you choose `swEndcondOffsetFromSurface` as the termination type for the first end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

TranslateSurface2

When you choose `swEndcondOffsetFromSurface` as the termination type for the second end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

NormalCut

True ensures that the cut is created normal to the sheet metal thickness, false does not (this argument applies only to sheet metal parts, use false for non-sheet metal parts)

UseFeatScope

True if the feature only affects selected bodies or components, false if the feature affects all bodies or components

UseAutoSelect

True to automatically select all bodies or components and have the feature affect those bodies or components, false to select the bodies or components the feature affects

AssemblyFeatureScope

True if the assembly feature only affects selected components in the assembly, false if the assembly feature affects all components in the assembly

AutoSelectComponents

True to auto-select all affected components, false to not (use the selected components only)

PropagateFeatureToParts

True to propagate the assembly feature to the components in the model that it affects, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.FeatureCut2](#).

Remarks

The default direction for cut operations are opposite the sketch normal. The default direction for boss operations is along the sketch normal. Setting the direction arguments to true reverse the default direction. For double-ended extrusions, Direction 2 is always be opposite to Direction 1.

The default sketch normal is the same as the face or plane normal where the sketch was placed. To determine this normal vector, see [IFace2::Normal](#) and [IRefPlane::Transform](#), respectively.

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IExtrudeFeatureData2 Interface](#)
[IFeatureManager::FeatureCutThicken Method](#)
[IFeatureManager::FeatureCutThin Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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FeatureCut3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCut3 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureCut4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCut3( _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean, _  
    ByVal TranslateSurface1 As System.Boolean, _  
    ByVal TranslateSurface2 As System.Boolean, _  
    ByVal NormalCut As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean, _  
    ByVal AssemblyFeatureScope As System.Boolean, _  
    ByVal AutoSelectComponents As System.Boolean, _  
    ByVal PropagateFeatureToParts As System.Boolean, _  
    ByVal T0 As System.Integer, _  
    ByVal StartOffset As System.Double, _  
    ByVal FlipStartOffset As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Sd As System.Boolean  
Dim Flip As System.Boolean  
Dim Dir As System.Boolean  
Dim T1 As System.Integer  
Dim T2 As System.Integer  
Dim D1 As System.Double  
Dim D2 As System.Double
```



```

Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim NormalCut As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim PropagateFeatureToParts As System.Boolean
Dim T0 As System.Integer
Dim StartOffset As System.Double
Dim FlipStartOffset As System.Boolean
Dim value As Feature

```

```

value = instance.FeatureCut3(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, NormalCut,
UseFeatScope, UseAutoSelect, AssemblyFeatureScope, AutoSelectComponents, PropagateFeatureToParts,
T0, StartOffset, FlipStartOffset)

```

C#

```

Feature FeatureCut3(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool NormalCut,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.bool AssemblyFeatureScope,
    System.bool AutoSelectComponents,
    System.bool PropagateFeatureToParts,
    System.int T0,
    System.double StartOffset,
    System.bool FlipStartOffset
)

```

C++/CLI

```

Feature^ FeatureCut3(
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,

```

```

& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool NormalCut,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents,
& System.bool PropagateFeatureToParts,
& System.int T0,
& System.double StartOffset,
& System.bool FlipStartOffset
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single-ended, false for doubled-ended

Flip

Flip side to cut; true if you want to remove material outside of the profile, false if not

Dir

Reverse direction; true if you want Direction 1 to be opposite of the default direction (see **Remarks**)

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

True for first draft angle to be inward, false to be outward

Ddir2

True for second draft angle to be inward, false be outward

Dang1

Draft angle for first end

Dang2

Draft angle for second end

OffsetReverse1

If you chose to offset the first end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

OffsetReverse2

If you chose to offset the second end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

TranslateSurface1

When you choose `swEndcondOffsetFromSurface` as the termination type for the first end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

TranslateSurface2

When you choose `swEndcondOffsetFromSurface` as the termination type for the second end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

NormalCut

True ensures that the cut is created normal to the sheet metal thickness, false does not (this argument applies only to sheet metal parts, use false for non-sheet metal parts)

UseFeatScope

True if the feature only affects selected bodies or components, false if the feature affects all bodies or components

UseAutoSelect

True to automatically select all bodies or components and have the feature affect those bodies or components, false to select the bodies or components the feature affects (see **Remarks**)

AssemblyFeatureScope

True if the assembly feature only affects selected components in the assembly, false if the assembly feature affects all components in the assembly

AutoSelectComponents

True to auto-select all affected components, false to not (use the selected components only)

PropagateFeatureToParts

True to propagate the assembly feature to the components in the model that it affects, false to not
T0
Start conditions as defined in [swStartConditions_e](#)
StartOffset
If T0 is swStartOffset, then specify an offset value
FlipStartOffset
If T0 is swStartOffset, then true to flip direction of cut, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCut3](#).

Example

[Insert Cut Extrude \(C#\)](#)
[Insert Cut Extrude \(VB.NET\)](#)
[Insert Cut Extrude \(VBA\)](#)

Remarks

The default direction for cut operations is opposite the sketch normal. The default direction for boss operations is along the sketch normal. Setting the direction arguments to true reverse the default direction. For double-ended extrusions, Direction 2 is always opposite to Direction 1.

The default sketch normal is the same as the face or plane normal where the sketch was placed. To determine this normal vector, see [IFace2::Normal](#) and [IRefPlane::Transform](#), respectively.

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IExtrudeFeatureData2 Interface](#)
[IFeatureManager::FeatureCutThin2 Method](#)
[IFeatureManager::FeatureCutThicken Method](#)

Availability

SOLIDWORKS 2009 SP4, Revision Number 17.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCut4 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCut4 Method (IFeatureManager)

Creates a cut extrude feature.

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCut4( _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _
```

```

ByVal OffsetReverse1 As System.Boolean, _
ByVal OffsetReverse2 As System.Boolean, _
ByVal TranslateSurface1 As System.Boolean, _
ByVal TranslateSurface2 As System.Boolean, _
ByVal NormalCut As System.Boolean, _
ByVal UseFeatScope As System.Boolean, _
ByVal UseAutoSelect As System.Boolean, _
ByVal AssemblyFeatureScope As System.Boolean, _
ByVal AutoSelectComponents As System.Boolean, _
ByVal PropagateFeatureToParts As System.Boolean, _
ByVal T0 As System.Integer, _
ByVal StartOffset As System.Double, _
ByVal FlipStartOffset As System.Boolean, _
ByVal OptimizeGeometry As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim NormalCut As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim PropagateFeatureToParts As System.Boolean
Dim T0 As System.Integer
Dim StartOffset As System.Double
Dim FlipStartOffset As System.Boolean
Dim OptimizeGeometry As System.Boolean
Dim value As Feature

value = instance.FeatureCut4(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, NormalCut,
UseFeatScope, UseAutoSelect, AssemblyFeatureScope, AutoSelectComponents, PropagateFeatureToParts,
T0, StartOffset, FlipStartOffset, OptimizeGeometry)

```

C#

```

Feature FeatureCut4(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,

```

```

System.bool OffsetReverse1,
System.bool OffsetReverse2,
System.bool TranslateSurface1,
System.bool TranslateSurface2,
System.bool NormalCut,
System.bool UseFeatScope,
System.bool UseAutoSelect,
System.bool AssemblyFeatureScope,
System.bool AutoSelectComponents,
System.bool PropagateFeatureToParts,
System.int T0,
System.double StartOffset,
System.bool FlipStartOffset,
System.bool OptimizeGeometry
)

```

C++/CLI

```

Feature^ FeatureCut4(
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool NormalCut,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents,
& System.bool PropagateFeatureToParts,
& System.int T0,
& System.double StartOffset,
& System.bool FlipStartOffset,
& System.bool OptimizeGeometry
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for a single-ended cut, false for a doubled-ended cut

Flip

True to remove material outside of the profile of the flip side to cut, false to not

Dir

True for Direction 1 to be opposite of the default direction (see **Remarks**)

T1

Termination type for the first end as defined in [swEndConditions_e](#)

T2

Termination type for the second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for the first end in meters

D2

Depth of extrusion for the second end in meters

Dchk1

True allows a draft angle in the first direction, false does not allow drafting in the first direction

Dchk2

True allows a draft angle in the second direction, false does not allow drafting in the second direction

Ddir1

True for the first draft angle to be inward, false to be outward; only valid when Dchk1 is true

Ddir2

True for the second draft angle to be inward, false to be outward; only valid when Dchk2 is true

Dang1

Draft angle for the first end; only valid when Dchk1 is true

Dang2

Draft angle for the second end; only valid when Dchk2 is true

OffsetReverse1

If you chose to offset the first end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

OffsetReverse2

If you chose to offset the second end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in a direction toward the sketch

TranslateSurface1

When you choose swEndConditions_e.swEndCondOffsetFromSurface as the termination type for the first end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

TranslateSurface2

When you choose swEndConditions_e.swEndCondOffsetFromSurface as the termination type for the second end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

NormalCut

True to create the cut normal to the sheet metal thickness, false to not (only valid for sheet metal parts; use false for non-sheet metal parts)

UseFeatScope

True if the feature only affects selected bodies or components, false if the feature affects all bodies or components

UseAutoSelect

True to automatically select all bodies or components and have the feature affect those bodies or components, false to only select the bodies or components the feature affects (see **Remarks**)

AssemblyFeatureScope

True if the assembly feature only affects selected components in the assembly, false if the assembly feature affects all components in the assembly

AutoSelectComponents

True to automatically select all affected components, false to use only the selected components

PropagateFeatureToParts

True to propagate the assembly feature to the components in the model that it affects, false to not

T0

Start conditions as defined in [swStartConditions_e](#)

StartOffset

If T0 is swStartConditions_e.swStartOffset, then specify an offset value

FlipStartOffset

If T0 is swStartConditions_e.swStartOffset, then true to flip the direction of cut, false to not

OptimizeGeometry

True to optimize the normal cut in a sheet metal part, false to not; only valid for sheet metal parts and when NormalCut is true

Return Value

Cut extrude [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCut4](#).

Example

[Create Cut Extrude in Sheet Metal Part \(C#\)](#)

[Create Cut Extrude in Sheet Metal Part \(VB.NET\)](#)

[Create Cut Extrude in Sheet Metal Part \(VBA\)](#)

Remarks

The default direction for:

- cut operations is opposite the sketch normal.
- boss operations is along the sketch normal.

Setting the direction parameters to true reverses the default direction. For double-ended extrusions, Direction 2 is always opposite to Direction 1

The default sketch normal is the face or plane that is perpendicular to the sketch plane. To determine the sketch normal, see [IFace2::Normal](#) and [IRefPlane::Transform](#).

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IExtrudeFeatureData2 Interface](#)

[IFeatureManager::FeatureCutThin2 Method \(\)](#)
[IFeatureManager::FeatureCutThicken Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCutThicken Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCutThicken Method (IFeatureManager)

Thickens the selected reference surface feature, and then generates a cut.

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCutThicken( _  
    ByVal Thickness As System.Double, _  
    ByVal Direction As System.Integer, _  
    ByVal FaceIndex As System.Integer, _  
    ByVal FillVolume As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Thickness As System.Double  
Dim Direction As System.Integer  
Dim FaceIndex As System.Integer  
Dim FillVolume As System.Boolean  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean  
Dim value As Feature
```

```
value = instance.FeatureCutThicken(Thickness, Direction, FaceIndex, FillVolume, UseFeatScope,  
UseAutoSelect)
```

C#

```
Feature FeatureCutThicken(  
    System.double Thickness,  
    System.int Direction,  
    System.int FaceIndex,  
    System.bool FillVolume,  
    System.bool UseFeatScope,  
    System.bool UseAutoSelect  
)
```

C++/CLI

```
Feature^ FeatureCutThicken(  
& System.double Thickness,  
& System.int Direction,  
& System.int FaceIndex,  
& System.bool FillVolume,  
& System.bool UseFeatScope,  
& System.bool UseAutoSelect  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness

Wall thickness

Direction

- 0 = Thicken side 1
- 1 = Thicken side 2
- 2 = Thicken both sides

FaceIndex

Not currently used

FillVolume

True to make the solid from a knitted surface, false to not (see **Remarks**)

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.FeatureCutThicken](#).

Remarks

This method creates a cut feature by thickening a selected reference surface. If FillVolume is True, other arguments are ignored. A closed surface is required when FillVolume is True.

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.FeatureCut2 Method](#)
[IFeatureManager.FeatureCutThin Method](#)
[IFeatureManager.FeatureBossThicken Method](#)
[IThickenFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureCutThin Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureCutThin Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.FeatureCutThin2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCutThin( _
    ByVal Sd As System.Boolean, _
    ByVal Flip As System.Boolean, _
    ByVal Dir As System.Boolean, _
    ByVal T1 As System.Integer, _
    ByVal T2 As System.Integer, _
    ByVal D1 As System.Double, _
    ByVal D2 As System.Double, _
    ByVal Dchk1 As System.Boolean, _
    ByVal Dchk2 As System.Boolean, _
    ByVal Ddir1 As System.Boolean, _
    ByVal Ddir2 As System.Boolean, _
    ByVal Dang1 As System.Double, _
    ByVal Dang2 As System.Double, _
```

```

ByVal OffsetReverse1 As System.Boolean, _
ByVal OffsetReverse2 As System.Boolean, _
ByVal TranslateSurface1 As System.Boolean, _
ByVal TranslateSurface2 As System.Boolean, _
ByVal Thk1 As System.Double, _
ByVal Thk2 As System.Double, _
ByVal EndThk As System.Double, _
ByVal RevThinDir As System.Integer, _
ByVal CapEnds As System.Integer, _
ByVal AddBends As System.Boolean, _
ByVal BendRad As System.Double, _
ByVal UseFeatScope As System.Boolean, _
ByVal UseAutoSelect As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim Thk1 As System.Double
Dim Thk2 As System.Double
Dim EndThk As System.Double
Dim RevThinDir As System.Integer
Dim CapEnds As System.Integer
Dim AddBends As System.Boolean
Dim BendRad As System.Double
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim value As Feature

```

```

value = instance.FeatureCutThin(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, Thk1, Thk2, EndThk,
RevThinDir, CapEnds, AddBends, BendRad, UseFeatScope, UseAutoSelect)

```

C#

```

Feature FeatureCutThin(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,

```

```

System.bool TranslateSurface2,
System.double Thk1,
System.double Thk2,
System.double EndThk,
System.int RevThinDir,
System.int CapEnds,
System.bool AddBends,
System.double BendRad,
System.bool UseFeatScope,
System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ FeatureCutThin(
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.double Thk1,
& System.double Thk2,
& System.double EndThk,
& System.int RevThinDir,
& System.int CapEnds,
& System.bool AddBends,
& System.double BendRad,
& System.bool UseFeatScope,
& System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single ended, false for double ended

Flip

Flip side to cut; true to remove material outside of the profile, false to not

Dir

Reverse direction; true if you want Direction1 to be opposite the default direction (see **Remarks**)

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

True for first draft angle to be inward, false to be outward

Ddir2

True for second draft angle to be inward, false to be outward

Dang1

First draft angle

Dang2

Second draft angle

OffsetReverse1

If you chose to offset the first end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in direction toward the sketch

OffsetReverse2

If you chose to offset the second end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in direction toward the sketch

TranslateSurface1

When you choose `swEndcondOffsetFromSurface` as the termination type for the first end, then `true` specifies that the end of the extrusion is a translation of the reference surface, `false` specifies to use a true offset

TranslateSurface2

When you choose `swEndcondOffsetFromSurface` as the termination type for the second end, then `true` specifies that the end of the extrusion is a translation of the reference surface, `false` specifies to use a true offset

Thk1

Wall thickness 1 (Mid-plane type uses $(Thk1)/2$ for each direction)

Thk2

Wall thickness 2 (only used when `RevThinDir=3`)

EndThk

End cap thickness (Only used when `CapEnds=1`)

RevThinDir

Thin feature type:

- 0 = One direction
- 1 = One direction reverse
- 2 = Mid-plane
- 3 = Two direction

CapEnds

Cap the ends:

- 0 = no cap
- 1 = cap (base features only)

AddBends

True to add auto bends (open profile base features only), `false` to not

BendRad

Fillet radii if `AddBends = True`

UseFeatScope

True if the feature only affects selected bodies, `false` if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, `false` to select the bodies the feature affects (see **Remarks**)

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCutThin](#).

Remarks

The default direction for cut operations is opposite the sketch normal. The default direction for boss operations is along the sketch normal. Setting the `Dir` argument to `True` reverses the default direction. For double-ended extrusions, `Direction 2` is always opposite of `Direction 1`.

The default sketch normal is the same as the face or plane normal where the sketch was placed. To determine this normal vector, see [IFace2::Normal](#) and [IRefPlane::Transform](#), respectively.

When `UseAutoSelect` is `false`, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IExtrudeFeatureData2 Interface](#)
[IFeatureManager::FeatureCut2 Method](#)
[IFeatureManager::FeatureCutThicken Method](#)
[IFeatureManager::FeatureExtrusionThin2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

Inserts a thin cut extrude feature.

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureCutThin2( _
    ByVal Sd As System.Boolean, _
    ByVal Flip As System.Boolean, _
    ByVal Dir As System.Boolean, _
    ByVal T1 As System.Integer, _
    ByVal T2 As System.Integer, _
    ByVal D1 As System.Double, _
    ByVal D2 As System.Double, _
    ByVal Dchk1 As System.Boolean, _
    ByVal Dchk2 As System.Boolean, _
    ByVal Ddir1 As System.Boolean, _
    ByVal Ddir2 As System.Boolean, _
    ByVal Dang1 As System.Double, _
    ByVal Dang2 As System.Double, _
    ByVal OffsetReverse1 As System.Boolean, _
    ByVal OffsetReverse2 As System.Boolean, _
    ByVal TranslateSurface1 As System.Boolean, _
    ByVal TranslateSurface2 As System.Boolean, _
    ByVal Thk1 As System.Double, _
    ByVal Thk2 As System.Double, _
    ByVal EndThk As System.Double, _
    ByVal RevThinDir As System.Integer, _
    ByVal CapEnds As System.Integer, _
    ByVal AddBends As System.Boolean, _
    ByVal BendRad As System.Double, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean, _
    ByVal T0 As System.Integer, _
    ByVal StartOffset As System.Double, _
    ByVal FlipStartOffset As System.Boolean _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim Thk1 As System.Double
Dim Thk2 As System.Double
Dim EndThk As System.Double
Dim RevThinDir As System.Integer
Dim CapEnds As System.Integer
Dim AddBends As System.Boolean
Dim BendRad As System.Double
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim T0 As System.Integer
```



```
Dim StartOffset As System.Double
Dim FlipStartOffset As System.Boolean
Dim value As Feature
```

```
value = instance.FeatureCutThin2(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, Thk1, Thk2, EndThk,
RevThinDir, CapEnds, AddBends, BendRad, UseFeatScope, UseAutoSelect, T0, StartOffset,
FlipStartOffset)
```

C#

```
Feature FeatureCutThin2(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.double Thk1,
    System.double Thk2,
    System.double EndThk,
    System.int RevThinDir,
    System.int CapEnds,
    System.bool AddBends,
    System.double BendRad,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.int T0,
    System.double StartOffset,
    System.bool FlipStartOffset
)
```

C++/CLI

```
Feature^ FeatureCutThin2(
    & System.bool Sd,
    & System.bool Flip,
    & System.bool Dir,
    & System.int T1,
    & System.int T2,
    & System.double D1,
    & System.double D2,
    & System.bool Dchk1,
    & System.bool Dchk2,
    & System.bool Ddir1,
    & System.bool Ddir2,
    & System.double Dang1,
    & System.double Dang2,
    & System.bool OffsetReverse1,
    & System.bool OffsetReverse2,
    & System.bool TranslateSurface1,
    & System.bool TranslateSurface2,
    & System.double Thk1,
    & System.double Thk2,
    & System.double EndThk,
    & System.int RevThinDir,
    & System.int CapEnds,
    & System.bool AddBends,
    & System.double BendRad,
```

```

& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.int T0,
& System.double StartOffset,
& System.bool FlipStartOffset
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single ended, false for double ended

Flip

Flip side to cut; true to remove material outside of the profile, false to not

Dir

Reverse direction; true if you want Direction1 to be opposite the default direction (see **Remarks**)

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

True for first draft angle to be inward, false to be outward

Ddir2

True for second draft angle to be inward, false to be outward

Dang1

First draft angle

Dang2

Second draft angle

OffsetReverse1

If you chose to offset the first end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in direction toward the sketch

OffsetReverse2

If you chose to offset the second end condition from another face or plane, then true specifies offset in direction away from the sketch, false specifies offset from the face or plane in direction toward the sketch

TranslateSurface1

When you choose swEndcondOffsetFromSurface as the termination type for the first end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

TranslateSurface2

When you choose swEndcondOffsetFromSurface as the termination type for the second end, then true specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

Thk1

Wall thickness 1 (mid-plane type uses $(Thk1)/2$ for each direction)

Thk2

Wall thickness 2 (only used when RevThinDir=3)

EndThk

End cap thickness (only used when CapEnds=1)

RevThinDir

Thin feature type:

- 0 = One direction
- 1 = One direction reverse
- 2 = Mid-plane
- 3 = Two direction

CapEnds

Cap the ends:

- 0 = no cap
- 1 = cap (base features only)

AddBends

True to add auto bends (open profile base features only), false to not

BendRad

Fillet radii if AddBends = True

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

T0

Start conditions as defined in [swStartConditions_e](#)

StartOffset

If T0 is swStartOffset, then specify an offset value

FlipStartOffset

If T0 is swStartOffset, then true to flip direction of cut, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureCutThin2](#).

Example

[Insert Thin Cut Extrude \(C#\)](#)
[Insert Thin Cut Extrude \(VB.NET\)](#)
[Insert Thin Cut Extrude \(VBA\)](#)

Remarks

The default direction for cut operations is opposite the sketch normal. The default direction for boss operations is along the sketch normal. Setting the Dir argument to True reverses the default direction. For double-ended extrusions, Direction 2 is always opposite of Direction 1.

The default sketch normal is the same as the face or plane normal where the sketch was placed. To determine this normal vector, see [IFace2::Normal](#) and [IRefPlane::Transform](#), respectively.

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IExtrudeFeatureData2 Interface](#)
[IFeatureManager::FeatureCut3 Method](#)
[IFeatureManager::FeatureCutThicken Method](#)
[IFeatureManager::FeatureExtrusionThin2 Method](#)

Availability

SOLIDWORKS 2009 SP4, Revision Number 17.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureDimensionPattern Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureDimensionPattern Method (IFeatureManager)

Not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureDimensionPattern( _  
    ByVal Num1 As System.Integer, _  
    ByVal Spacing1 As System.Double, _  
    ByVal Num2 As System.Integer, _  
    ByVal Spacing2 As System.Double, _  
    ByVal DiagonalOnly As System.Boolean, _  
    ByVal DName1 As System.String, _  
    ByVal DName2 As System.String, _  
    ByVal VaryInstance As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Num1 As System.Integer  
Dim Spacing1 As System.Double  
Dim Num2 As System.Integer  
Dim Spacing2 As System.Double  
Dim DiagonalOnly As System.Boolean
```



```
Dim DName1 As System.String
Dim DName2 As System.String
Dim VaryInstance As System.Boolean
Dim value As Feature
```

```
value = instance.FeatureDimensionPattern(Num1, Spacing1, Num2, Spacing2, DiagonalOnly, DName1, DName2, VaryInstance)
```

C#

```
Feature FeatureDimensionPattern(
    System.int Num1,
    System.double Spacing1,
    System.int Num2,
    System.double Spacing2,
    System.bool DiagonalOnly,
    System.string DName1,
    System.string DName2,
    System.bool VaryInstance
)
```

C++/CLI

```
Feature^ FeatureDimensionPattern(
    & System.int Num1,
    & System.double Spacing1,
    & System.int Num2,
    & System.double Spacing2,
    & System.bool DiagonalOnly,
    & System.String^ DName1,
    & System.String^ DName2,
    & System.bool VaryInstance
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Num1

Spacing1

Num2

Spacing2

DiagonalOnly

DName1

DName2

VaryInstance

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

Availability

Not implemented

SOLIDWORKS API Help

[Send comments](#) on this topic.

FeatureExtruRefSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureExtruRefSurface Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureExtruRefSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub FeatureExtruRefSurface( _
    ByVal Sd As System.Boolean, _
    ByVal Flip As System.Boolean, _
    ByVal Dir As System.Boolean, _
    ByVal T1 As System.Integer, _
    ByVal T2 As System.Integer, _
    ByVal D1 As System.Double, _
    ByVal D2 As System.Double, _
    ByVal Dchk1 As System.Boolean, _
    ByVal Dchk2 As System.Boolean, _
    ByVal Ddir1 As System.Boolean, _
    ByVal Ddir2 As System.Boolean, _
    ByVal Dang1 As System.Double, _
    ByVal Dang2 As System.Double, _
    ByVal OffsetReverse1 As System.Boolean, _
    ByVal OffsetReverse2 As System.Boolean, _
    ByVal TranslateSurface1 As System.Boolean, _
    ByVal TranslateSurface2 As System.Boolean, _
    ByVal CapEnd1 As System.Boolean, _
    ByVal CapEnd2 As System.Boolean, _
    ByVal DeleteOriginalFace As System.Boolean, _
    ByVal KnitResult As System.Boolean _
)

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim CapEnd1 As System.Boolean
Dim CapEnd2 As System.Boolean
Dim DeleteOriginalFace As System.Boolean
Dim KnitResult As System.Boolean

```

```

instance.FeatureExtruRefSurface(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, CapEnd1, CapEnd2,
DeleteOriginalFace, KnitResult)

```

C#

```

void FeatureExtruRefSurface(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,

```

```

    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool CapEnd1,
    System.bool CapEnd2,
    System.bool DeleteOriginalFace,
    System.bool KnitResult
)

```

C++/CLI

```

void FeatureExtruRefSurface(
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool CapEnd1,
& System.bool CapEnd2,
& System.bool DeleteOriginalFace,
& System.bool KnitResult
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single-ended, false for double-ended

Flip

True to flip the direction to cut, false to not

Dir

True to flip direction of the extrude, false to not

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

For first draft angle to be inward use true, for draft angle outward use false

Ddir2

For second draft angle to be inward use true, for draft angle outward use false

Dang1

Draft angle for first end

Dang2

Draft angle for second end

OffsetReverse1

If you choose to offset the first end condition from another face or plane, then true specifies the offset from the face or plane in a direction away from the sketch, and false specifies the offset in a direction toward the sketch

OffsetReverse2

If you choose to offset the second end condition from another face or plane, then true specifies the offset from the face or plane in direction away from the sketch, and false specifies the offset in a direction toward the sketch

TranslateSurface1

If you specified T1 as [swEndConditions](#) .swEndCondOffsetFromSurface, then true specifies that the end of the extrusion is a translation of the reference surface, and false specifies to use a true offset

TranslateSurface2

If you specified T2 as [swEndConditions](#) .swEndCondOffsetFromSurface, then true specifies that the end of the extrusion is a translation of the reference surface, and false specifies to use a true offset

CapEnd1

True to cap first end, false to not

CapEnd2

True to cap second end, false to not

DeleteOriginalFace

True to delete original faces, false to not (see **Remarks**)

KnitResult

True to knit the extrusion result, false to not (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureExtruRefSurface](#).

Remarks

Before calling this method, select the entities from which to extrude reference surfaces. Do one of the following:

- select a sketch containing one face or multiple co-planar faces
- select one face or multiple co-planar faces in the view

DeleteOriginalFace and KnitResult are ignored if the extrusion is in two directions.

KnitResult is ignored if DeleteOriginalFace is false.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureExtruRefSurface2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureExtruRefSurface2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureExtruRefSurface3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub FeatureExtruRefSurface2( _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean, _  
    ByVal TranslateSurface1 As System.Boolean, _  
    ByVal TranslateSurface2 As System.Boolean, _
```

```

ByVal CapEnd1 As System.Boolean, _
ByVal CapEnd2 As System.Boolean, _
ByVal DeleteOriginalFace As System.Boolean, _
ByVal KnitResult As System.Boolean _
)

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim CapEnd1 As System.Boolean
Dim CapEnd2 As System.Boolean
Dim DeleteOriginalFace As System.Boolean
Dim KnitResult As System.Boolean

```

```

instance.FeatureExtruRefSurface2(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1,
Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, CapEnd1, CapEnd2,
DeleteOriginalFace, KnitResult)

```

C#

```

void FeatureExtruRefSurface2(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool CapEnd1,
    System.bool CapEnd2,
    System.bool DeleteOriginalFace,
    System.bool KnitResult
)

```

C++/CLI

```

void FeatureExtruRefSurface2(
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,

```

```

& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool CapEnd1,
& System.bool CapEnd2,
& System.bool DeleteOriginalFace,
& System.bool KnitResult
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single-ended, false for double-ended

Flip

Not used

Dir

True to reverse the direction of the extrude, false to not

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

True for first draft angle to be outward, false for first draft angle to be inward

Ddir2

True for second draft angle to be outward, false for second draft angle to be inward

Dang1

Draft angle for first end

Dang2

Draft angle for second end

OffsetReverse1

If you choose to offset the first end condition from another face or plane, then true specifies the offset from the face or plane in a direction away from the sketch, and false specifies the offset in a direction toward the sketch

OffsetReverse2

If you choose to offset the second end condition from another face or plane, then true specifies the offset from the face or plane in a direction away from the sketch, and false specifies the offset in a direction toward the sketch

TranslateSurface1

If you specified T1 as [swEndConditions_e.swEndCondOffsetFromSurface](#), then true specifies that the end of the extrusion is a translation of the reference surface, and false specifies to use a true offset

TranslateSurface2

If you specified T2 as [swEndConditions_e.swEndCondOffsetFromSurface](#), then true specifies that the end of the extrusion is a translation of the reference surface, and false specifies to use a true offset

CapEnd1

True to cap first end, false to not

CapEnd2

True to cap second end, false to not

DeleteOriginalFace

True to delete original faces, false to not (see **Remarks**)

KnitResult

True to knit the extrusion result, false to not (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureExtruRefSurface2](#).

Example

[Create Trimmed Surface Feature \(C#\)](#)
[Create Trimmed Surface Feature \(VB.NET\)](#)
[Create Trimmed Surface Feature \(VBA\)](#)

Remarks

Before calling this method:

- Select the base of the extrusion surface:
 - Sketch; call [IModelDocExtension::SelectByID2](#) with selection mark 0.
 - Sketch segments; call [IModelDocExtension::SelectByID2](#) with selection mark 0 or 4.
 - Single planar face; call [IModelDocExtension::SelectByID2](#) with selection mark 0.
 - Multiple co-planar faces; call [IModelDocExtension::SelectByID2](#) with selection mark 0 for the first planar face and selection mark 256 for other co-planar faces.
- Select the entity reference for the direction of the extrusion by calling [IModelDocExtension::SelectByID2](#) with selection mark 16. If no entity is selected, the direction of extrusion is normal to the selected extrusion surface.

DeleteOriginalFace and KnitResult are ignored if the extrusion is in two directions.

KnitResult is ignored if DeleteOriginalFace is false.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfExtrudeFeatureData Interface](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

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FeatureExtruRefSurface3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureExtruRefSurface3 Method (IFeatureManager)

Inserts an extruded surface.

.NET Syntax

Visual Basic (Declaration)

```
Sub FeatureExtruRefSurface3( _  
    ByVal Sd As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal StartCond As System.Integer, _  
    ByVal OffsetVal As System.Double, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean, _  
    ByVal TranslateSurface1 As System.Boolean, _  
    ByVal TranslateSurface2 As System.Boolean, _  
    ByVal CapEnd1 As System.Boolean, _  
    ByVal CapEnd2 As System.Boolean, _  
    ByVal DeleteOriginalFace As System.Boolean, _  
    ByVal KnitResult As System.Boolean _  
)
```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Dir As System.Boolean
Dim StartCond As System.Integer
Dim OffsetVal As System.Double
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim CapEnd1 As System.Boolean
Dim CapEnd2 As System.Boolean
Dim DeleteOriginalFace As System.Boolean
Dim KnitResult As System.Boolean

```

```

instance.FeatureExtruRefSurface3(Sd, Dir, StartCond, OffsetVal, T1, T2, D1, D2, Dchk1, Dchk2,
Ddir1, Ddir2, Dang1, Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2,
CapEnd1, CapEnd2, DeleteOriginalFace, KnitResult)

```

C#

```

void FeatureExtruRefSurface3(
    System.bool Sd,
    System.bool Dir,
    System.int StartCond,
    System.double OffsetVal,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool CapEnd1,
    System.bool CapEnd2,
    System.bool DeleteOriginalFace,
    System.bool KnitResult
)

```

C++/CLI

```

void FeatureExtruRefSurface3(
& System.bool Sd,
& System.bool Dir,
& System.int StartCond,
& System.double OffsetVal,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,

```

```

& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool CapEnd1,
& System.bool CapEnd2,
& System.bool DeleteOriginalFace,
& System.bool KnitResult
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single-ended, false for double-ended

Dir

True to reverse the direction of the extrude, false to not

StartCond

Start condition as defined in [swStartConditions_e](#)

OffsetVal

Offset value

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#) (valid only if *Sd* = false)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters (valid only if *Sd* = false)

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting (valid only if *Sd* = false)

Ddir1

True for first draft angle to be outward, false for first draft angle to be inward

Ddir2

True for second draft angle to be outward, false for second draft angle to be inward (valid only if *Sd* = false)

Dang1

Draft angle for first end

Dang2

Draft angle for second end (valid only if *Sd* = false)

OffsetReverse1

If you choose to offset the first end condition from another face or plane, then true specifies the offset from the face or plane in a direction away from the sketch, and false specifies the offset in a direction toward the sketch

OffsetReverse2

If you choose to offset the second end condition from another face or plane, then true specifies the offset from the face or plane in a direction away from the sketch, and false specifies the offset in a direction toward the sketch (valid only if *Sd* = false)

TranslateSurface1

If you specified *T1* as [swEndConditions_e.swEndCondOffsetFromSurface](#), then true specifies that the end of the extrusion is a translation of the reference surface, and false specifies to use a true offset

TranslateSurface2

If you specified *T2* as [swEndConditions_e.swEndCondOffsetFromSurface](#), then true specifies that the end of the extrusion is a translation of the reference surface, and false specifies to use a true offset (valid only if *Sd* = false)

CapEnd1

True to cap first end, false to not

CapEnd2

True to cap second end, false to not (valid only if *Sd* = false)

DeleteOriginalFace

True to delete original faces, false to not (valid only if *Sd* = true)

KnitResult

True to knit the extrusion result, false to not (valid only if *Sd* = true and *DeleteOriginalFace* = true)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.FeatureExtruRefSurface3](#).

Example

[Insert Extruded Surface \(C#\)](#)
[Insert Extruded Surface \(VB.NET\)](#)

Remarks

Before calling this method, select the entities for the extruded surface using [IModelDocExtension::SelectByID2](#).

Entity	Mark
Base	
• 2D or 3D sketch • Sketch segments • Single planar face • Multiple co-planar faces	• 0 • 0 or 4 • 0 • 0 for first planar face and 256 for all other co-planar faces
Direction edge	16
Start condition entity	32
End condition entity	1

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfExtrudeFeatureData Interface](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

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FeatureExtrusion Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureExtrusion Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureExtrusion2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureExtrusion( _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean, _  
    ByVal TranslateSurface1 As System.Boolean, _  
    ByVal TranslateSurface2 As System.Boolean, _  
    ByVal Merge As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Sd As System.Boolean  
Dim Flip As System.Boolean  
Dim Dir As System.Boolean  
Dim T1 As System.Integer  
Dim T2 As System.Integer  
Dim D1 As System.Double
```

```

Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim Merge As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim value As Feature

```

```

value = instance.FeatureExtrusion(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2,
Dang1, Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, Merge,
UseFeatScope, UseAutoSelect)

```

C#

```

Feature FeatureExtrusion(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool Merge,
    System.bool UseFeatScope,
    System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ FeatureExtrusion(
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool Merge,
& System.bool UseFeatScope,
& System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

- Sd*
- Flip*
- Dir*
- T1*
- T2*
- D1*
- D2*
- Dchk1*
- Dchk2*
- Ddir1*
- Ddir2*
- Dang1*
- Dang2*
- OffsetReverse1*
- OffsetReverse2*
- TranslateSurface1*
- TranslateSurface2*
- Merge*
- UseFeatScope*
- UseAutoSelect*

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureExtrusion](#).

 See Also

- [IFeatureManager Interface](#)
- [IFeatureManager Members](#)

FeatureExtrusion2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : FeatureExtrusion2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureExtrusion3](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function FeatureExtrusion2( _
    ByVal Sd As System.Boolean, _
    ByVal Flip As System.Boolean, _
    ByVal Dir As System.Boolean, _
    ByVal T1 As System.Integer, _
    ByVal T2 As System.Integer, _
    ByVal D1 As System.Double, _
    ByVal D2 As System.Double, _
    ByVal Dchk1 As System.Boolean, _
    ByVal Dchk2 As System.Boolean, _
    ByVal Ddir1 As System.Boolean, _
    ByVal Ddir2 As System.Boolean, _
    ByVal Dang1 As System.Double, _
    ByVal Dang2 As System.Double, _
    ByVal OffsetReverse1 As System.Boolean, _
    ByVal OffsetReverse2 As System.Boolean, _
    ByVal TranslateSurface1 As System.Boolean, _
    ByVal TranslateSurface2 As System.Boolean, _
```

```

    ByVal Merge As System.Boolean, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean, _
    ByVal T0 As System.Integer, _
    ByVal StartOffset As System.Double, _
    ByVal FlipStartOffset As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean
Dim Merge As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim T0 As System.Integer
Dim StartOffset As System.Double
Dim FlipStartOffset As System.Boolean
Dim value As Feature

```

```

value = instance.FeatureExtrusion2(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2,
Dang1, Dang2, OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2, Merge,
UseFeatScope, UseAutoSelect, T0, StartOffset, FlipStartOffset)

```

C#

```

Feature FeatureExtrusion2(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2,
    System.bool Merge,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.int T0,
    System.double StartOffset,
    System.bool FlipStartOffset
)

```

C++/CLI

```

Feature^ FeatureExtrusion2(
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2,
& System.bool TranslateSurface1,
& System.bool TranslateSurface2,
& System.bool Merge,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.int T0,
& System.double StartOffset,
& System.bool FlipStartOffset
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

True for single ended, false for double ended

Flip

True to flip side to cut

Dir

True to flip the direction to extrude

T1

Termination type for first end as defined in [swEndConditions_e](#)

T2

Termination type for second end as defined in [swEndConditions_e](#)

D1

Depth of extrusion for first end in meters

D2

Depth of extrusion for second end in meters

Dchk1

True allows draft angle in first direction, false does not allow drafting

Dchk2

True allows draft angle in second direction, false does not allow drafting

Ddir1

True for first draft angle to be inward, false to be outward

Ddir2

True for second draft angle to be inward, false to be outward

Dang1

Draft angle for first end

Dang2

Draft angle for second end

OffsetReverse1

If you chose to offset the first end condition from another face or plane, then True specifies offset in direction away from the sketch, false specifies offset from the face or plane in direction toward the sketch

OffsetReverse2

If you chose to offset the second end condition from another face or plane, then True specifies offset in direction away from the sketch, false specifies offset from the face or plane in direction toward the sketch

TranslateSurface1

When you choose swEndcondOffsetFromSurface as the termination type for the first end, then True specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

TranslateSurface2

When you choose swEndcondOffsetFromSurface as the termination type for the second end, then True specifies that the end of the extrusion is a translation of the reference surface, false specifies to use a true offset

Merge

True to merge the results in a multibody part, false to not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

t0

Start condition as defined in [swStartConditions_e](#)

StartOffset

If t0 set to swStartOffset, then specify offset value

FlipStartOffset

If t0 set to swStartOffset, then True to flip the direction or false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::FeatureExtrusion2](#).

Remarks

The default direction for cut operations is opposite the sketch normal. The default direction for boss operations is along the sketch normal. Setting the Dir argument to True reverses the default direction. For double-ended extrusions, Direction 2 is always opposite of Direction 1.

The default sketch normal is the same as the face or plane normal where the sketch was placed. To determine this normal vector, see [IFace2::Normal](#) and [IRefPlane::Transform](#), respectively.

When useAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

To extrude a 3D sketch, select:

- 3D sketch with selection mark 0.
- Extrusion direction edge with selection mark 16.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IFeatureManager::FeatureExtrusionThin2 Method](#)

[IFeatureManager::FeatureCut Method](#)

[IExtrudeFeatureData2 Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0